

Worksheet 2.1

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1. Aim/Overview of the practical: To implement zero-shot and few-shot prompting techniques using a Large Language Model through the text generation tasks.

2. Coding:

```
from sambanova import SambaNova
```

```
client = SambaNova(  
    api_key=".....",  
    base_url="https://api.sambanova.ai/v1",  
)
```

```
# ZERO SHOT TEXT GENERATION
```

```
response1 = client.chat.completions.create(  
    model="Meta-Llama-3.3-70B-Instruct",  
    messages=[  
        {"role": "system", "content": "You are a helpful assistant."},  
        {"role": "user", "content": "Write a short paragraph on the importance of Generative AI."}  
    ],  
    temperature=0.7,  
)  
print("\nZERO SHOT TEXT GENERATION:\n")  
print(response1.choices[0].message.content)
```

```
# ZERO SHOT CLASSIFICATION
```

```
response2 = client.chat.completions.create(  
    model="Meta-Llama-3.3-70B-Instruct",  
    messages=[  
        {"role": "system", "content": "You are a sentiment classifier."},  
        {"role": "user", "content": ""}  
    ]  
    )  
Classify the sentiment as Positive, Negative, or Neutral.  
Text: "The lab session was very interesting and informative."
```



```
    }],  
    temperature=0,  
)  
print("\nZERO SHOT CLASSIFICATION:\n")  
print(response2.choices[0].message.content)  
  
# FEW SHOT CLASSIFICATION  
  
response3 = client.chat.completions.create(  
    model="Meta-Llama-3.3-70B-Instruct",  
    messages=[  
        {"role": "system", "content": "You are a sentiment classifier."},  
        {"role": "user", "content": ""}  
        Classify sentiment:  
        Text: I love this subject - Positive  
        Text: This assignment is terrible - Negative  
        Text: The exam is tomorrow - Neutral  
        Now classify:  
        Text: The teacher explained everything clearly -  
        ""  
    ],  
    temperature=0,  
)  
print("\nFEW SHOT CLASSIFICATION:\n")  
print(response3.choices[0].message.content)
```

3. OUTPUT:

ZERO SHOT TEXT GENERATION:

Generative AI has emerged as a revolutionary technology with far-reaching implications, transforming the way we create, innovate, and interact. By enabling machines to generate new content, such as images, videos, music, and text, generative AI has the potential to unlock unprecedented levels of creativity, productivity, and efficiency. Its importance lies in its ability to automate repetitive tasks, enhance human capabilities, and facilitate novel solutions to complex problems. Furthermore, generative AI has numerous applications across industries, including art, design, healthcare, and education, making it an essential tool for driving progress, fostering innovation, and improving overall quality of life. As the technology continues to evolve, its significance is likely to grow, opening up new avenues for exploration, discovery, and advancement.



ZERO SHOT CLASSIFICATION:

The sentiment of the text is: Positive.

FEW SHOT CLASSIFICATION:

The sentiment of the text "The teacher explained everything clearly" is: Positive.

The text expresses a favorable opinion about the teacher's explanation, indicating satisfaction and approval.

4. Learning outcomes :

- Generate coherent text outputs by providing only task instructions (zero-shot).
- Classify sentiment without examples (zero-shot) and with examples (few-shot).
- Map numerical predictions to human-readable categories (Negative, Neutral, Positive).
- Demonstrate zero-shot and few-shot prompting techniques using a large language model.
- How to use and integrate different generated API keys.